EE227 – Digital Logic Design

Lecture 19 – 20

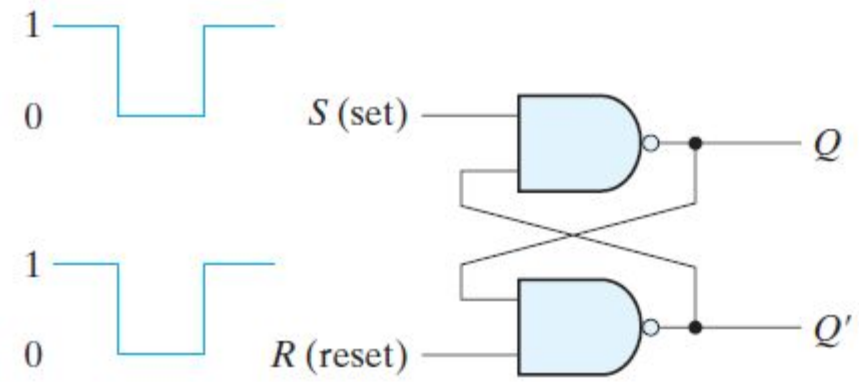
Outline

- Flip Flops
- D – Flip Flop
- JK Flip Flop
- T – Flip Flop
- Quiz 3 Solution
- Code Conversion Combinational Circuit
- Quiz 4
- Quiz 4 Solution

Latch and Flip Flop

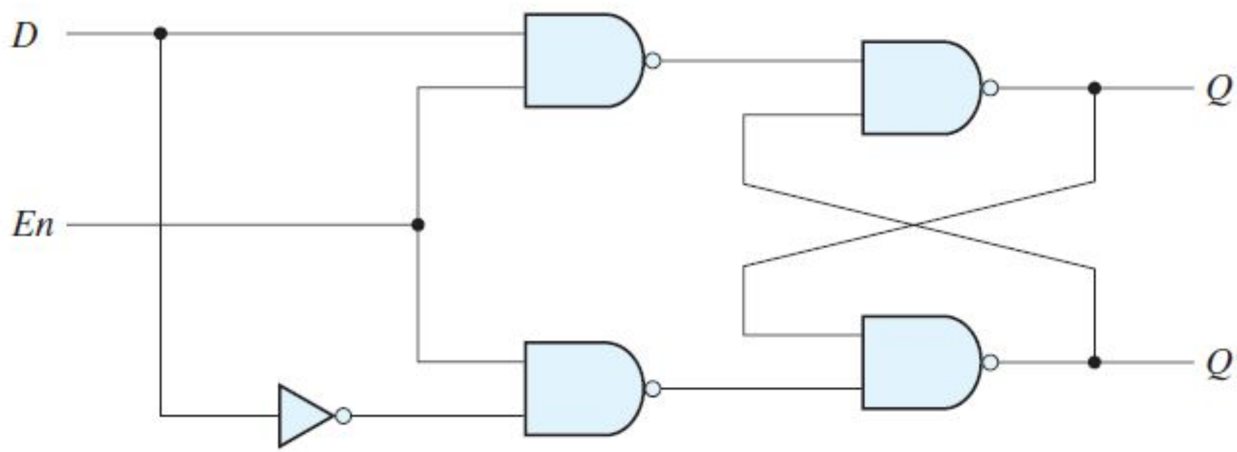
- Latches
  - Storage elements that operate with signal levels are referred to as latches
- Flip Flops
  - Storage elements that are controlled by a clock transition are flip-flops

SR Latch



S	R	Q	State
0	0	?	Forbidden
0	1	1	Set
1	0	0	Reset
1	1	Previous State	No Change

D Latch



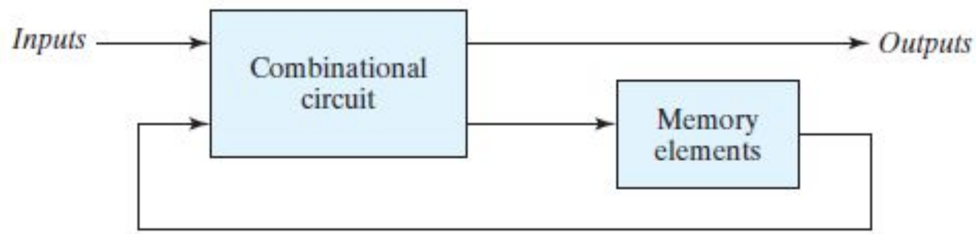
En	D	Next State of Q
0	X	No Change
1	0	Q = 0 → Reset State
1	1	Q = 1 → Set State

Conversion of D Latch to Flip Flop

- The state of a latch or flip-flop is switched by a change in the control input
- This momentary change is called a trigger, and the transition it causes is said to trigger the flip-flop
- The D latch with pulses in its control input is essentially a flip-flop that is triggered every time the pulse goes to the logic-1 level
- As long as the pulse input remains at this level, any changes in the data input will change the output and the state of the latch

Problem with using Latches in Sequential Circuits (1/2)

- In a sequential circuit the inputs of the flip-flops are derived in part from the outputs of the same and other flip-flops



- The state transitions of the latches start as soon as the clock pulse changes to the logic-1 level
- The new state of a latch appears at the output while the pulse is still active
- If the inputs applied to the latches change while the clock pulse is still at the logic-1 level, the latches will respond to new values and a new output state may occur
- The result is an unpredictable situation

Problem with using Latches in Sequential Circuits (2/2)

- If the inputs applied to the latches change while the clock pulse is still at the logic-1 level, the latches will respond to new values and a new output state may occur
- The result is an unpredictable situation
- Because of this unreliable operation, the output of a latch cannot be applied directly or through combinational logic to the input of the same or another latch when all the latches are triggered by a common clock source

Flip Flops (1/3)

- Flip-flop circuits are constructed in such a way as to make them operate properly when they are part of a sequential circuit that employs a common clock
- The problem with the latch is that it responds to a change in the level of a clock pulse

Response to positive level

Flip Flops (2/3)

- The key to the proper operation of a flip-flop is to trigger it only during a signal transition
- A clock pulse goes through two transitions:
  - From 0 to 1
    - Positive edge transition

Positive-edge response

  - From 1 to 0
    - Negative edge transition

Negative-edge response

Flip Flops (3/3)

- There are two ways that a latch can be modified to form a flip-flop
  - One way is to employ two latches in a special configuration that isolates the output of the flip-flop and prevents it from being affected while the input to the flip-flop is changing
  - Another way is to produce a flip-flop that triggers only during a signal transition (from 0 to 1 or from 1 to 0) of the synchronizing signal (clock) and is disabled during the rest of the clock pulse

Edge Triggered D – Flip Flop with D Latches

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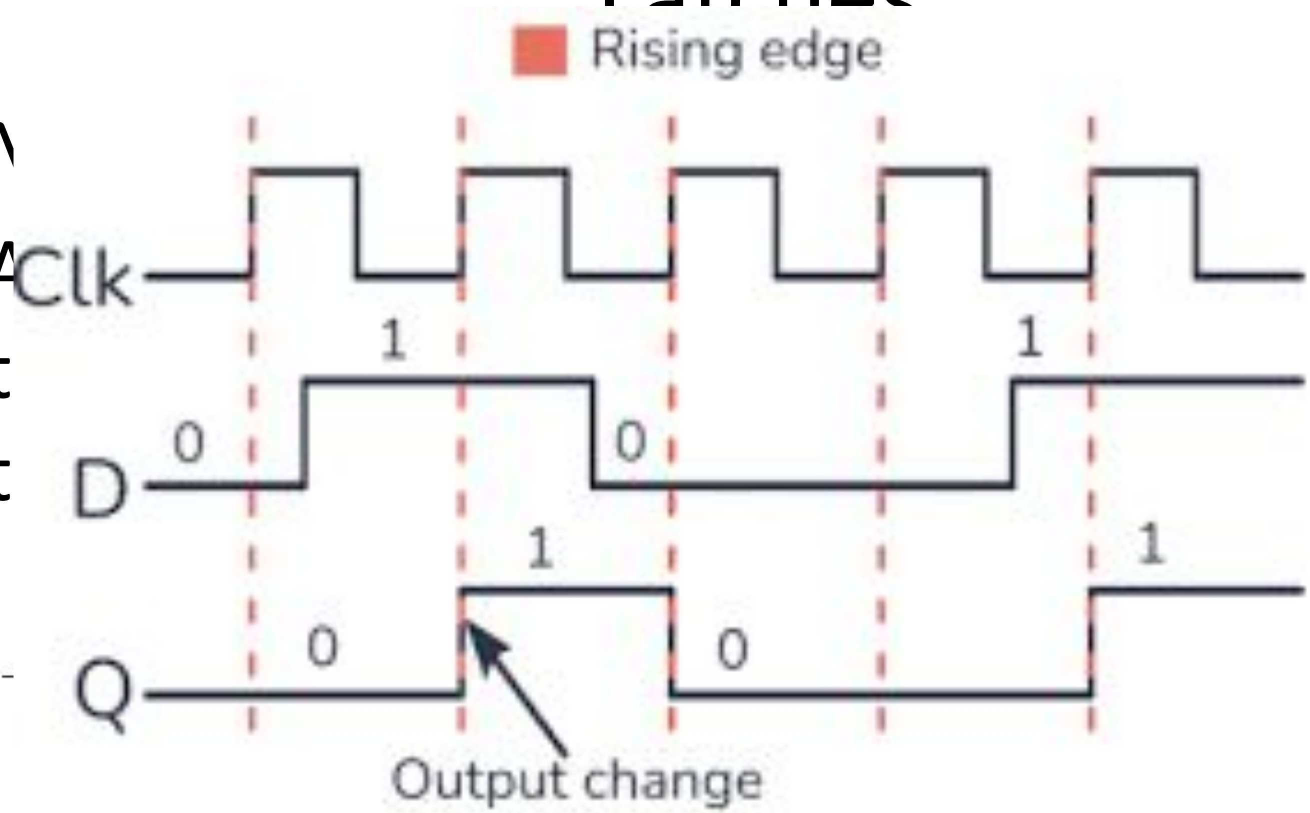
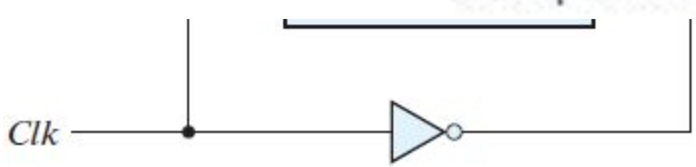
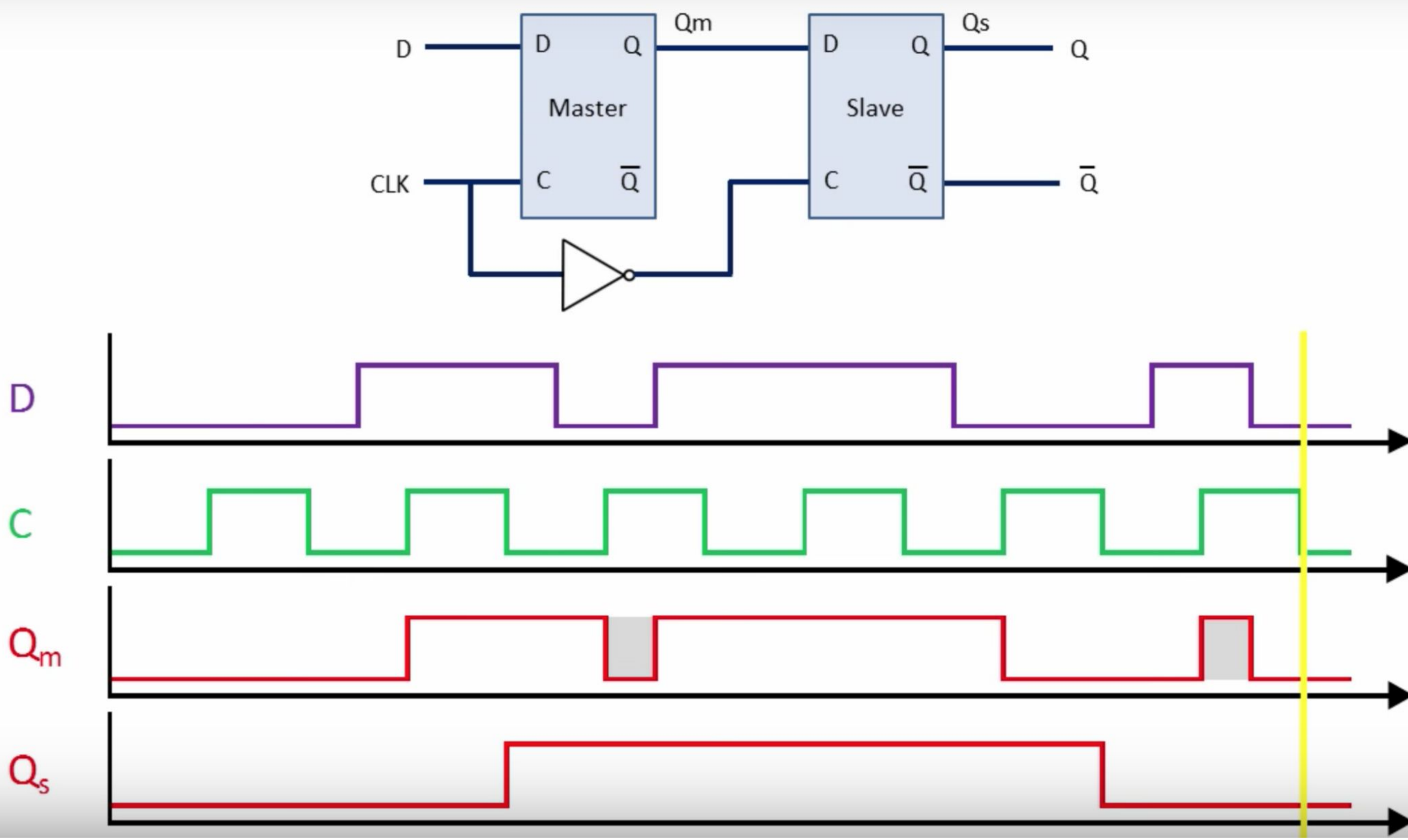
- A ... can be t ... tion of t





	$Q(t+1)$	
	0	Reset
1	1	Set

Edge Triggered D – Flip Flop with SR

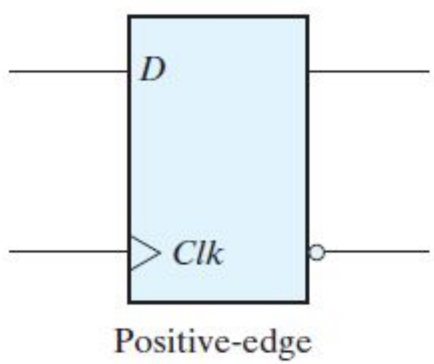


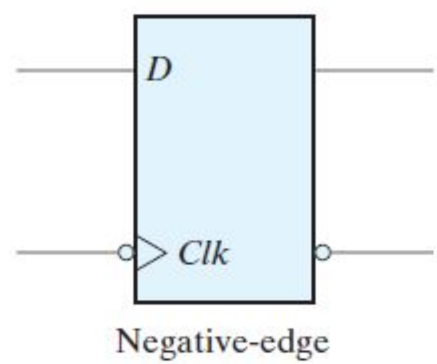
Setup, Hold and Propagation Delay Time

- There is a minimum time called the **setup time** during which the D input must be maintained at a constant value prior to the occurrence of the clock transition
- There is a minimum time called the **hold time** during which the D input must not change after the application of the positive transition of the clock
- The **propagation delay time** of the flip-flop is defined as the interval between the trigger edge and the stabilization of the output to a new state

Graphical Symbols of D – Flip Flop

- The dynamic indicator (>) denotes the fact that the flip-flop responds to the edge transition of the clock



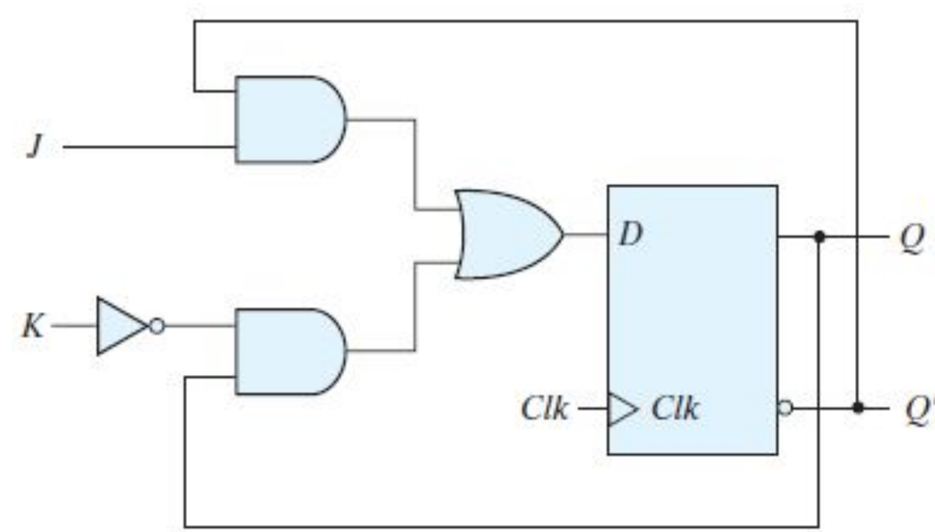


Other Flip Flops

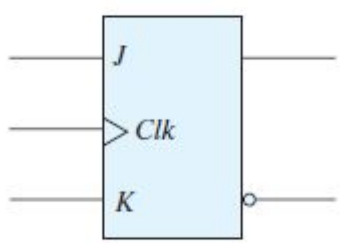
- There are three operations that can be performed with a flip-flop
  - Set it to 1
  - Reset it to 0
  - Complement its output
- With only a single input, the D flip-flop can set or reset the output
- There are two other flip flops
  - JK flip flop
  - T flip flop

JK Flip Flop

- 2 inputs J and K
- Performs all 3 operations
- D = JQ' + K'Q



J	K	$Q(t+1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

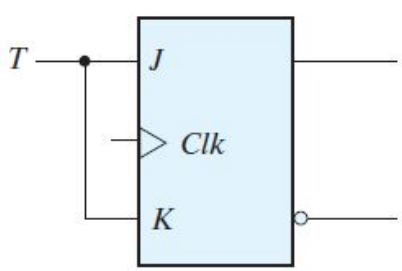


Toggle (T) – Flip Flop

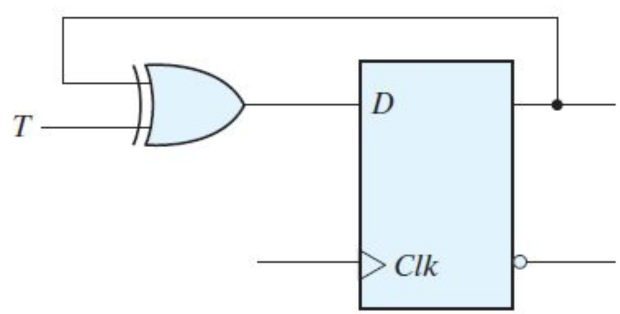
- The T (toggle) flip-flop is a complementing flip-flop and can be obtained from a JK flip-flop when inputs J and K are tied together
- The T flip-flop can be constructed with a D flip-flop and an exclusive-OR gate as

$$D = T \oplus Q = TQ' + T'Q$$

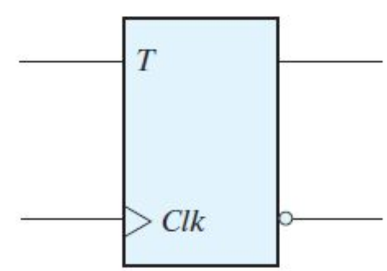
T	$Q(t+1)$	
0	$Q(t)$	No change
1	$Q'(t)$	Complement



From *JK* flip-flop



From *D* flip-flop



Graphic symbol

Characteristic Equations

- The next state [Q(t + 1)] can be obtained with the help of characteristic equations
- For D – Flip Flop

$$Q(t + 1) = D$$

- For JK Flip Flop

$$Q(t + 1) = JQ' + K'Q$$

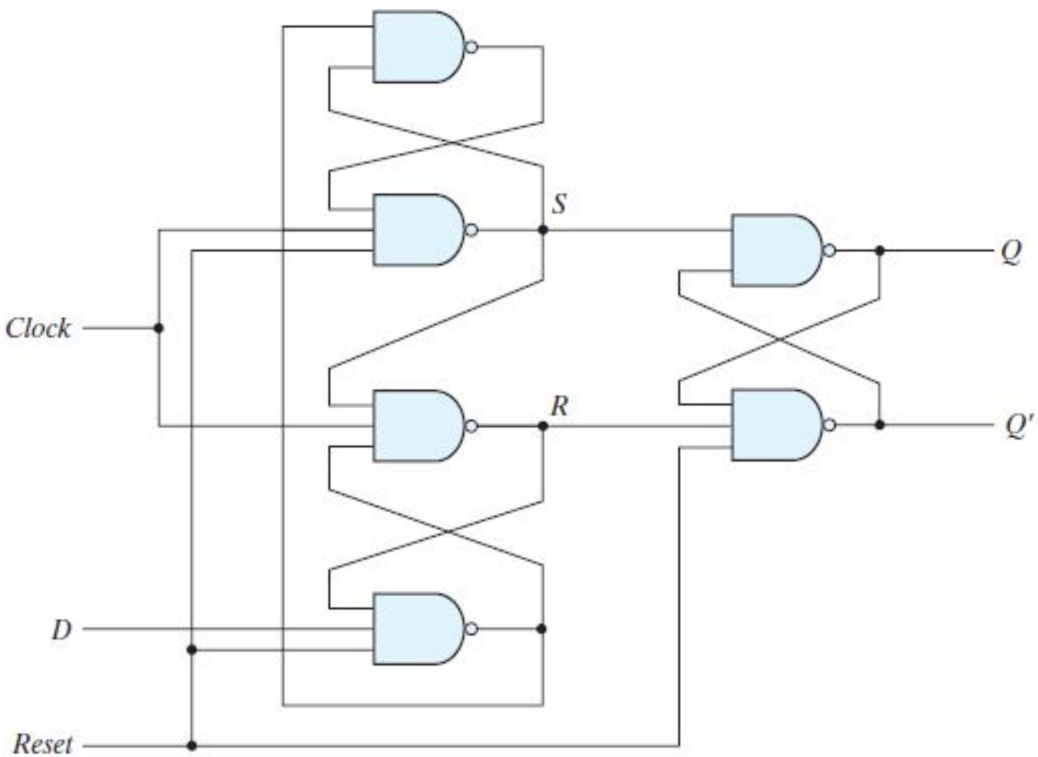
- For T – Flip Flop

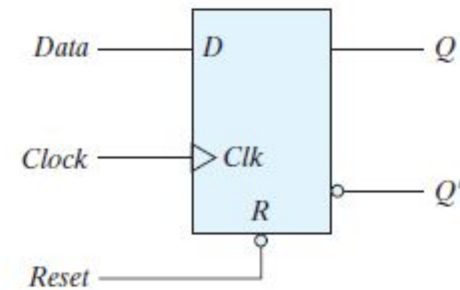
$$Q(t + 1) = T \oplus Q = TQ' + T'Q$$

Direct Inputs in Flip Flops

- Some flip-flops have asynchronous inputs that are used to force the flip-flop to a particular state independently of the clock
- The input that sets the flip-flop to 1 is called **preset** or **direct set**
- The input that clears the flip-flop to 0 is called **clear** or **direct reset**

Edge Triggered D Flip Flop with Direct Reset





R	Clk	D	Q	Q'
0	X	X	0	1
1	↑	0	0	1
1	↑	1	1	0

Code Conversion Combinational Circuit

Missed Topic from Chapter 4

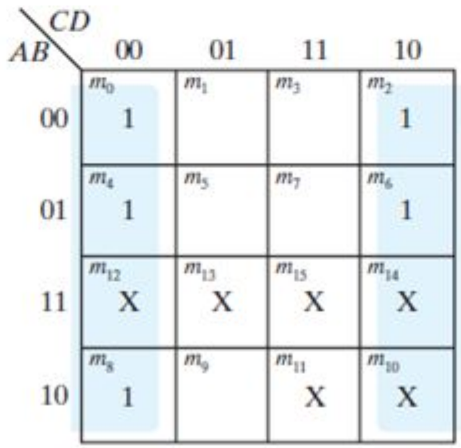
BCD to Excess – 3 Code Conversion (1/4)

- Truth Table

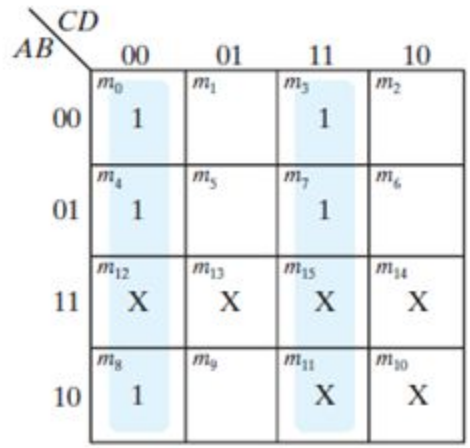
Inputs				Outputs			
A	**B**	**C**	**D**	**w**	**x**	**y**	**z**
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
1	0	1	0	X	X	X	X
1	0	1	1	X	X	X	X
1	1	0	0	X	X	X	X
1	1	0	1	X	X	X	X
1	1	1	0	X	X	X	X
1	1	1	1	X	X	X	X

BCD to Excess – 3 Code Conversion (2/4)

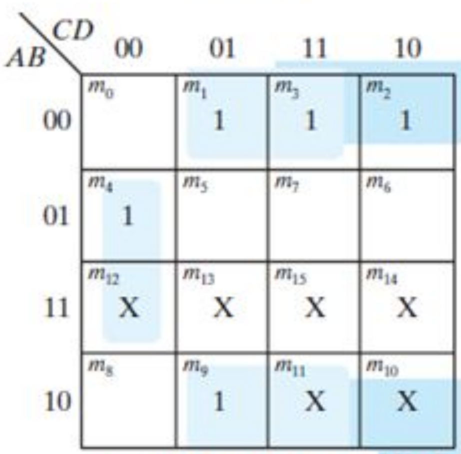
- K – Map Simplification



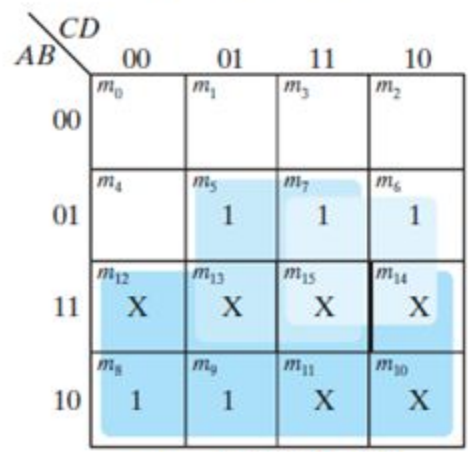
$z = D'$



$y = CD + C'D'$



$x = B'C + B'D + BC'D'$



$w = A + BC + BD$

BCD to Excess – 3 Code Conversion (3/4)

- Boolean Equations

$z = D'$

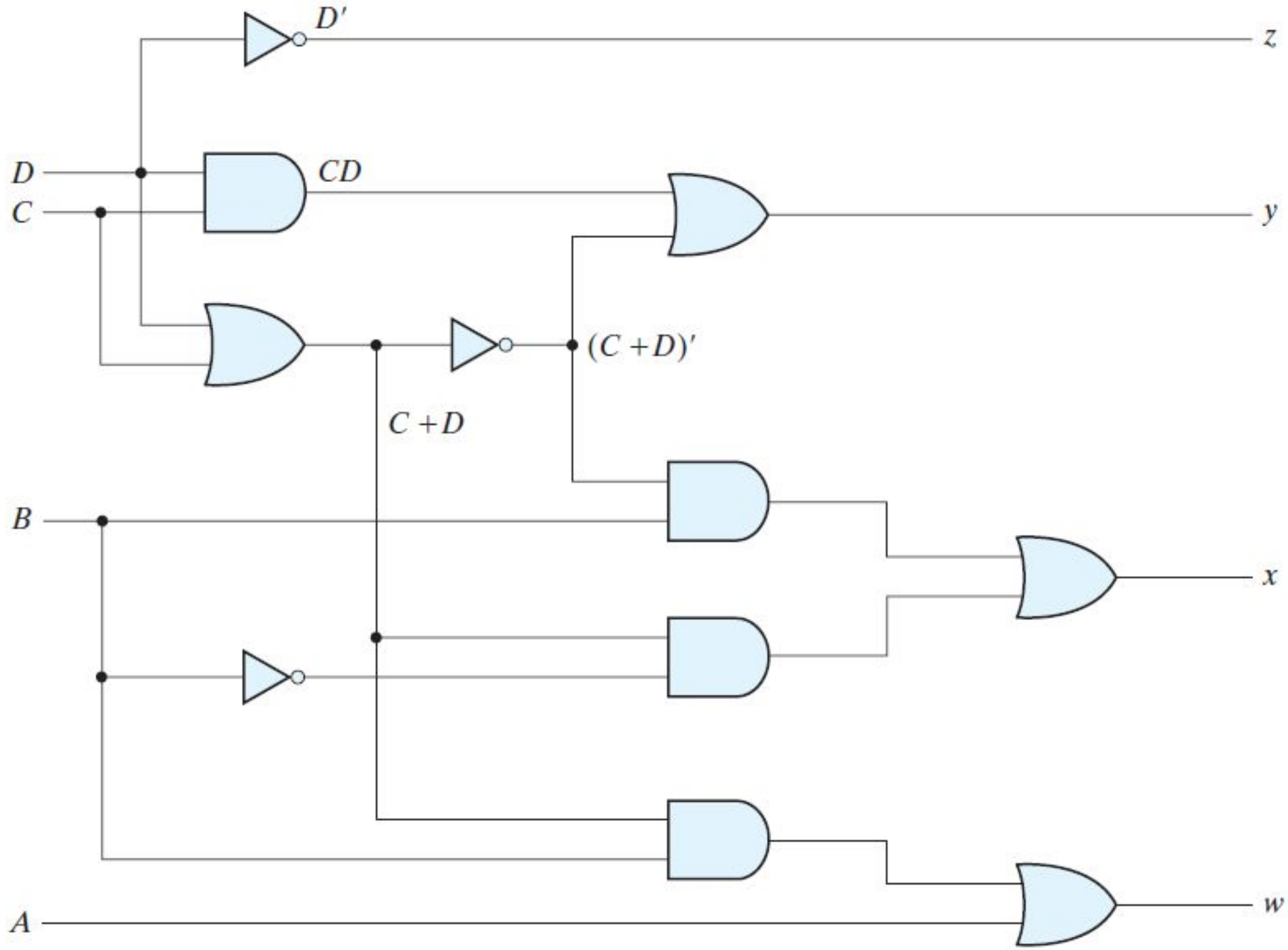
$y = CD + C'D'$
$y = CD + (C + D)'$

$x = B'C + B'D + BC'D'$
$x = B'(C + D) + BC'D'$
$x = B'(C + D) + B(C + D)'$

$w = A + BC + BD$
$w = A + B(C + D)$

BCD to Excess – 3 Code Conversion (4/4)

- Circuit Diagram



Quiz 3 and 4

Solution

Was Done in Class